

Generalized Sampling of Non-Local Textural Clues Multi-View Stereo Framework

Jingyuan Tang*

tangjingyuan2018@bupt.edu.cn

the School of Information and Communication
Engineering, Beijing University of Posts and Telecommu-
nications (BUPT)
Key Laboratory of Trustworthy Distributed Computing
and Service (BUPT), Ministry of Education
Beijing, China

Xuesong Gao

State Key Laboratory of Digital Multimedia Technology,
Hisense Co., Ltd.
Qingdao, China
College of Intelligence and Computing, Tianjin University
Tianjin, China
xuesong@outlook.com

Yangang Cai*†

the School of Information and Communication
Engineering, Beijing University of Posts and Telecommu-
nications (BUPT)
Beijing, China
caiyangang@bupt.edu.cn

Songlin Sun

the School of Information and Communication
Engineering, Beijing University of Posts and Telecommu-
nications (BUPT)
Key Laboratory of Trustworthy Distributed Computing
and Service (BUPT), Ministry of Education
Beijing, China
slsun@bupt.edu.cn

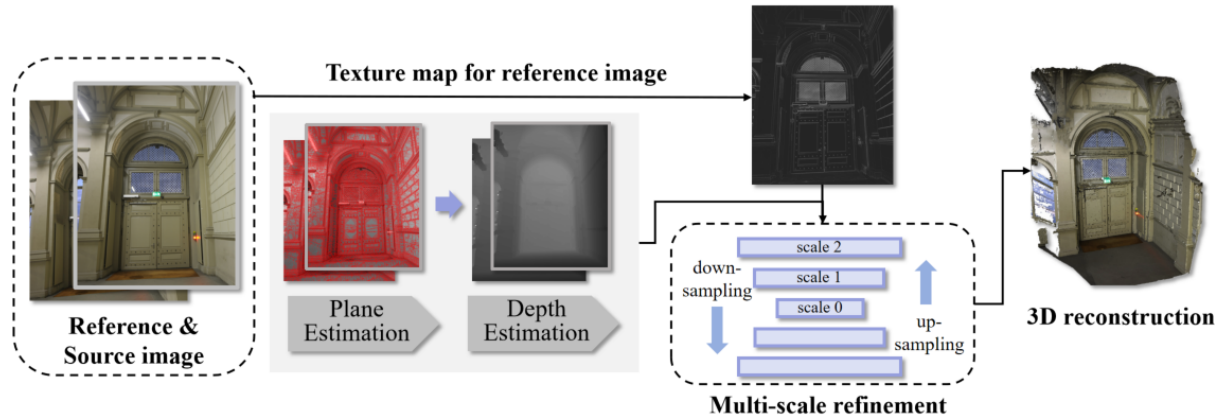


Figure 1: Architecture of GST-MVS.

Abstract

3D reconstruction remains a pivotal topic in contemporary multimedia research, with multi-view stereo (MVS) methods being instrumental. Traditional MVS algorithms, have demonstrated robust performance in large-scale scenes. However, these methods

often struggle with depth estimation in non-textured regions, resulting in inconsistencies across different areas of the reconstructed 3D model. This paper introduces a novel MVS framework, General Sampling Non-local Texture Features (GST-MVS), designed to enhance depth estimation by leveraging the intrinsic relationship between image texture and object depth. Our approach optimizes the 3D reconstruction process, yielding superior results, especially in low-textured regions. Comprehensive evaluations on the ETH3D dataset indicate that GST-MVS achieves dense and accurate 3D point cloud reconstructions, outperforming existing MVS algorithms. The GST-MVS framework is open-source and accessible on GitHub at <https://github.com/Jasmine-tjy/GST-mvs>. Additionally, Docker image facilitating the required execution environment, which is available at https://hub.docker.com/r/tangjas111/cmake_cuda_opencv/tags.

* Both authors contributed equally to this research.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.
MM '24, October 28–November 1, 2024, Melbourne, VIC, Australia.

© 2024 Copyright held by the owner/author(s). Publication rights licensed to ACM.
ACM ISBN 979-8-4007-0686-8/24/10
<https://doi.org/10.1145/3664647.3685516>

CCS Concepts

• Computing methodologies → Point-based models.

Keywords

Point cloud reconstruction, Multi-view stereo, 3D reconstruction

ACM Reference Format:

Jingyuan Tang, Yangang Cai, Xuesong Gao, and Songlin Sun. 2024. Generalized Sampling of Non-Local Textural Clues Multi-View Stereo Framework. In *Proceedings of the 32nd ACM International Conference on Multimedia (MM '24)*, October 28–November 1, 2024, Melbourne, VIC, Australia. ACM, New York, NY, USA, 4 pages. <https://doi.org/10.1145/3664647.3685516>

1 Introduction

In the past few years, 3D reconstruction has attracted widespread attention as an important task in computer vision and has made many progress. MVS is a fundamental task in computer vision, involving the construction of dense 3D models from a collection of multi-view images and their corresponding camera parameters. MVS technology finds extensive application across various fields, including virtual reality, autonomous driving, remote sensing, and more. Currently, mainstream MVS processes typically involve predicting the depth map for each input perspective and utilizing these depth maps to generate a 3D point cloud model, thereby achieving a balance between efficiency and flexibility. Xu [9] has optimized based on the PatchMatch [1] and has become a phased optimization anchor. In recent years, MVS algorithms have largely been further optimized based on it. MVS algorithm primarily focuses on two types of methods: traditional approaches and learning-based methods. Learning-based methods leverage deep learning networks to extract high-level features, aiming to enhance the prediction accuracy across the four stages of the PatchMatch pipeline. However, learning-based methods often necessitate considerable storage due to the iteration of numerous feature vectors throughout intermediate steps.

Traditional MVS methods generally follow two mainstream approaches: planar scanning based and PatchMatch based. Due to its superior performance in extensive scenes, PatchMatch remains one of the optimal solutions for dense reconstruction tasks. This method involves traversing a set of images along with the corresponding camera parameters, randomly initializing the plane assumption for each pixel in the image, and then calculating the confidence level of this plane assumption based on photometric consistency. PatchMatch effectively estimates the depth of each pixel by transforming the depth estimation problem into a plane estimation task, evaluating the confidence level, and propagating the plane assumption of the reliable image. 3D point clouds can be densely reconstructed using this method. However, due to the reliance of PatchMatch based MVS methods on photometric consistency for depth estimation, significant depth estimation errors can occur in non-textured and low-textured areas, resulting in decreased accuracy and completeness. Therefore, enhancing the reliability of depth estimation in textureless regions presents a significant challenge for this method.

Our proposed framework provides targeted solutions to the aforementioned issues. We build upon the concept of multi-scale depth estimation and enhance the overall quality of 3D reconstruction by

employing generalized sampling of non-local texture features. The design of our work has the following advantages:

- We leverage the relationship between texture and depth to adaptively expand the sampling scheme using texture constraints. This approach broadens the receptive field and reduces the occurrence of outliers. Notably, this scheme **can be seamlessly integrated** into most existing PatchMatch based MVS methods.
- To mitigate the negative impact of low-texture regions on photometric consistency measurement, we incorporate texture factors into the normalized cross-correlation measurement, coupled with an adaptive extension process. This helps to improve the integrity of the reconstructed scene.

Finally, we have developed a PatchMatch based MVS method that has demonstrated excellent performance on the ETH3D dataset [4] comparing to other method. Regarding the completeness of 3D reconstruction, the framework demonstrated an improvement of approximately 3% within a 2 cm accuracy range.

2 Architecture of GST-MVS

2.1 Multi Scale Image Pyramid Structure

The multi-scale image pyramid structure extracts image features from different receptive fields for each perspective by processing the image at different scales. Due to the tendency of depth estimation in low texture areas using Patchmatch based methods, pyramid structures can generate and optimize depth maps for processing low texture areas, resulting in more accurate depth estimation. The specific steps of GST-MVS are as follows:

- **Generating Initial Depth Maps**
At the coarsest scale, we initially utilize the plane assumption in triangulation to derive initial depth maps for all images, along with corresponding texture maps for reference images. This step serves as the foundation for subsequent optimization, providing a preliminary estimate of depth.
- **Upsampling and Detail Refinement**
The depth and texture maps of all images are upsampled to the next finer scale. This upsampling process allows reliable depth estimates and texture values from low texture regions to propagate to the current scale. At each new scale, plane estimation is employed to refine details and correct errors inherited from the previous scale.
- **Optimization of Plane Assumptions at the Same Geometry Scale**
At the current scale, the corrected depth and texture maps are used as initializations. A plane prior estimate, consistent with geometric features, is applied again. This optimization enhances depth estimation reliability in low texture regions and ensures consistency across different images.
- **Iterative Refinement**
The formal steps are iteratively repeated at progressively finer scales until the original image resolution is achieved. Each iteration improves the depth map's accuracy, leading to a highly refined final depth map.

2.2 General Sampling Non-local Texture Features

The PatchMatch-based method approaches depth estimation by treating it as plane estimation. It calculates the matching cost for a plane using a patch from the reference image and its corresponding patch from the source image. View weights are aggregated from multiple sources, and if the cost is minimal, the depth value propagates to the pixel. Texture mapping refines this process, and in low-texture areas, the continuity of pixels and point clouds in 2D and 3D allows for expanded propagation, improving the 3D reconstruction's integrity.

Specifically, during the sampling process, the pixel sampling range in non-textured and low-textured areas can be adaptively expanded based on the degree of texture, thereby enlarging the sampling window. A larger window encompasses a wider receptive field, enabling more reliable estimation of planar assumptions in these regions. The matching cost quantifies the correspondence between pixels in different perspective source images, with its magnitude reflecting the stability of pixels across perspectives. We analyze the cost and texture levels of pixels in various views, tally the values within favorable and unfavorable ranges, and determine the necessity of adaptive expansion through multiple iterations. This iterative process ensures the robustness and accuracy of the depth estimation.

In order to achieve precise expansion and improve the accuracy of matching, we adopted an adaptive expansion sampling strategy in 8 directions. For instance, if a preliminary decision expands the current pixel upwards, sampling the pixels above and comparing cost values in different directions identifies the position with the lowest cost. The multi-view cost calculation method is then selected based on this position. If it is above the edge area, the multi-view cost vector is recalculated; if within the edge region, the previously calculated cost vector is inherited. Through this approach, we can adaptively expand sampling in different directions. In addition, to address the adverse effects of low-texture regions on photometric consistency measurements, we integrate texture factors into the normalized cross-correlation (NCC) framework and employ an adaptive extension process. This approach enhances the integrity and accuracy of the reconstructed scene by ensuring more robust photometric consistency, particularly in areas with low texture.

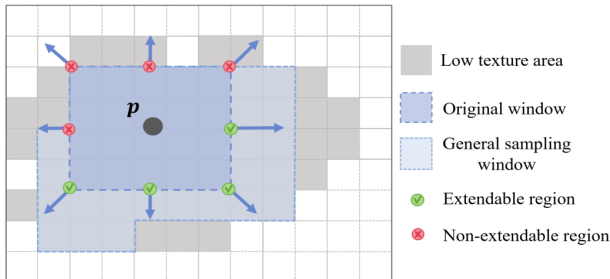


Figure 2: Visualization for computation of GST algorithm.

Figure 2 is a visual representation of the extended sampling process, where gray areas represent textureless or low texture areas,

dark blue areas represent the original sampling window, and light blue areas represent the adaptive extended sampling area.

3 Demonstrations and Experiment

Running the GST-MVS 3D point cloud reconstruction software involves three essential steps.

- Standardizing the Dataset

The dataset provided on the ETH3D official website has been sparsely reconstructed using Colmap. This image dataset has been calibrated with a set of undistorted camera extrinsic parameters and view selection. Prior to executing the code, it is imperative to standardize the dataset and export the camera parameters and intermediate data. The intermediate data encompasses the external parameters of the image, the corresponding information between the two-dimensional features of the image and the three-dimensional spatial points, the coordinates of the three-dimensional spatial points in the world coordinate system, RGB values, and trajectories on various images.

- Pulling the Docker Image and Running the Container

Due to version compatibility issues among CUDA, OpenCV, and CMake, setting up the construction environment is quite complex. To simplify this process, we have provided a tailored Docker image for this model, which can be accessed and pulled from https://hub.docker.com/r/tangjas111/cmake_cuda_opencv/tags.

- Compiling and Executing the Code

Compile and run the code according to the following command to achieve 3D reconstruction.

```
cmake .
make
./GST <ETH3D root path >
```

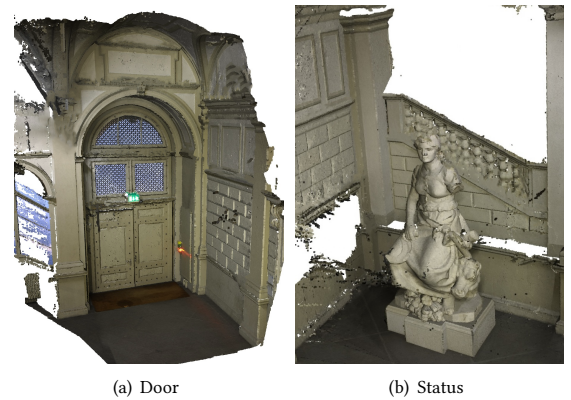


Figure 3: 3D reconstruction instance from GST-MVS

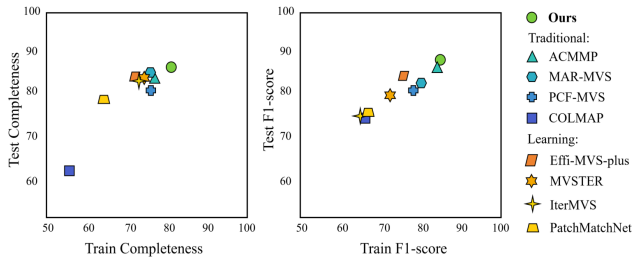
GST-MVS facilitates the rapid and efficient reconstruction of 3D point clouds, even for small-scale scenes. Accurate scene reconstruction can be achieved with a limited number of images. As demonstrated in Figures 3(a) and 3(b), the door was reconstructed

Table 1: Point cloud evaluation on ETH3D benchmark. We show completeness / accuracy / F1 score (in %) at different thresholds (2cm and 10cm).

	Method	2cm			10cm		
Train	(i) <i>Traditional</i>						
	COLMAP[3]	55.13	91.85	67.66	79.47	98.75	87.61
	PCF-MVS[2]	75.73	84.11	79.42	90.42	95.98	92.98
	MAR-MVS[10]	77.19	81.98	79.21	90.44	95.51	92.71
	ACMMP[9]	77.61	<u>90.63</u>	83.42	93.32	<u>97.99</u>	95.54
	(ii) <i>Learning</i>						
	PatchMatchNet[6]	65.43	64.81	64.21	83.28	89.98	85.70
	IterMVS[5]	61.87	73.62	66.36	78.85	94.48	85.25
	MVSTER[8]	76.92	68.08	72.06	91.91	91.97	91.73
	Effi-MVS-plus[7]	70.42	82.33	75.28	-	-	-
	Ours	80.25	89.98	84.71	94.48	97.79	96.06
Test	(i) <i>Traditional</i>						
	COLMAP[3]	62.98	91.97	73.01	84.54	98.25	90.4
	PCF-MVS[2]	79.29	82.15	80.38	91.26	92.12	91.56
	MAR-MVS[10]	84.18	80.24	81.84	94.43	94.21	94.22
	ACMMP[9]	81.49	<u>91.91</u>	85.89	94.67	<u>98.05</u>	96.27
	(ii) <i>Learning</i>						
	PatchMatchNet[6]	77.46	69.71	73.12	92.05	91.98	91.91
	IterMVS[5]	72.65	76.91	74.29	85.81	95.42	90.15
	MVSTER[8]	82.47	77.09	79.01	92.71	94.21	92.30
	Effi-MVS-plus[7]	83.81	83.90	83.62	-	-	-
	Ours	84.63	90.96	87.42	95.97	97.82	96.85

with a high degree of completeness using only seven images. Similarly, the statue was successfully reconstructed using just eleven images.

We conducted extensive testing on the publicly available ETH3D dataset to evaluate the performance of our method against recently released 3D reconstruction techniques. The evaluation was based on three key metrics: completeness, accuracy, and the F1 score. The results of this comparative analysis are presented in Table 1. In the table, the best results are highlighted in bold, while the second-best results are underlined to facilitate a clear comparison. For a more intuitive understanding, a graphical performance of completeness and F1 score comparison is provided in Figure 4.

**Figure 4: Comparison with the learning-based methods and traditional methods on ETH3D dataset.**

The above results illustrate that our method demonstrates superior performance in MVS 3D reconstruction. It maintains high robustness and accuracy even in challenging scenarios, such as

significant lighting variations or areas with insufficient texture information.

4 Conclusion and Outlook

In this study, we have introduced generalized sampling of non-Local textural clues method for 3D reconstruction, which substantially enhances the completeness of reconstructed objects. Our method achieves this improvement by adaptively expanding the range of propagation depth estimation, thereby facilitating more accurate and detailed reconstructions. To ensure ease of use and reproducibility, we have provided a Docker image, which simplifies the environmental setup for users. Additionally, we have developed a comprehensive demonstration to guide users through the detailed process of running the code. This versatile framework enables the 3D reconstruction of objects in both small and large scenes using only a few images. Such adaptability renders our method highly practical for a wide array of applications, ranging from small-scale object reconstruction to extensive environmental mapping.

We are committed to the ongoing development and refinement of the GST-MVS model. Future efforts will focus on further enhancing the completeness and accuracy of reconstructions, which includes improving the model's capacity to manage complex scenes and variable image qualities, as well as optimizing computational efficiency.

References

- [1] Michael Bleyer, Christoph Rhemann, and Carsten Rother. 2011. Patchmatch stereo-stereo matching with slanted support windows.. In *Bmvc*, Vol. 11. 1–11.
- [2] Andreas Kuhn, Shan Lin, and Oliver Erdler. 2019. Plane Completion and Filtering for Multi-View Stereo Reconstruction. In *Pattern Recognition*, Gernot A. Fink, Simone Frintrop, and Xiaoyi Jiang (Eds.). Springer International Publishing, Cham, 18–32.
- [3] Johannes L Schönberger, Enliang Zheng, Jan-Michael Frahm, and Marc Pollefeys. 2016. Pixelwise view selection for unstructured multi-view stereo. In *Computer Vision–ECCV 2016: 14th European Conference, Amsterdam, The Netherlands, October 11–14, 2016, Proceedings, Part III 14*. Springer, 501–518.
- [4] Thomas Schöps, Johannes L. Schönberger, Silvano Galliani, Torsten Sattler, Konrad Schindler, Marc Pollefeys, and Andreas Geiger. 2017. A Multi-view Stereo Benchmark with High-Resolution Images and Multi-camera Videos. In *2017 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*. 2538–2547. <https://doi.org/10.1109/CVPR.2017.272>
- [5] Fangjinhua Wang, Silvano Galliani, Christoph Vogel, and Marc Pollefeys. 2022. IterMVS: Iterative Probability Estimation for Efficient Multi-View Stereo. In *2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 8596–8605. <https://doi.org/10.1109/CVPR52688.2022.00841>
- [6] Fangjinhua Wang, Silvano Galliani, Christoph Vogel, Pablo Speciale, and Marc Pollefeys. 2021. PatchmatchNet: Learned Multi-View Patchmatch Stereo. In *2021 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 14189–14198. <https://doi.org/10.1109/CVPR46437.2021.01397>
- [7] Shaoqian Wang, Bo Li, and Yuchao Dai. 2024. Efficient Multi-view Stereo by Dynamic Cost Volume and Cross-scale Propagation. *IEEE Transactions on Circuits and Systems for Video Technology* (2024), 1–1. <https://doi.org/10.1109/TCSVT.2024.3398060>
- [8] Xiaofeng Wang, Zheng Zhu, Guan Huang, Fangbo Qin, Yun Ye, Yijia He, Xu Chi, and Xingang Wang. 2022. MVSTER: Epipolar Transformer for Efficient Large Scale Scene Reconstruction. In *Computer Vision – ECCV 2022: 17th European Conference, Tel Aviv, Israel, October 23–27, 2022, Proceedings, Part XXXI* (Tel Aviv, Israel). Springer-Verlag, Berlin, Heidelberg, 573–591. https://doi.org/10.1007/978-3-031-19821-2_33
- [9] Qingshan Xu, Weihang Kong, Wenbing Tao, and Marc Pollefeys. 2023. Multi-Scale Geometric Consistency Guided and Planar Prior Assisted Multi-View Stereo. *IEEE Transactions on Pattern Analysis and Machine Intelligence* 45, 4 (2023), 4945–4963. <https://doi.org/10.1109/TPAMI.2022.3200074>
- [10] Zhenyu Xu, Yiguang Liu, Xuelei Shi, Ying Wang, and Yunan Zheng. 2020. MAR-MVS: Matching Ambiguity Reduced Multiple View Stereo for Efficient Large Scale Scene Reconstruction. In *2020 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 5980–5989. <https://doi.org/10.1109/CVPR42600.2020.00602>